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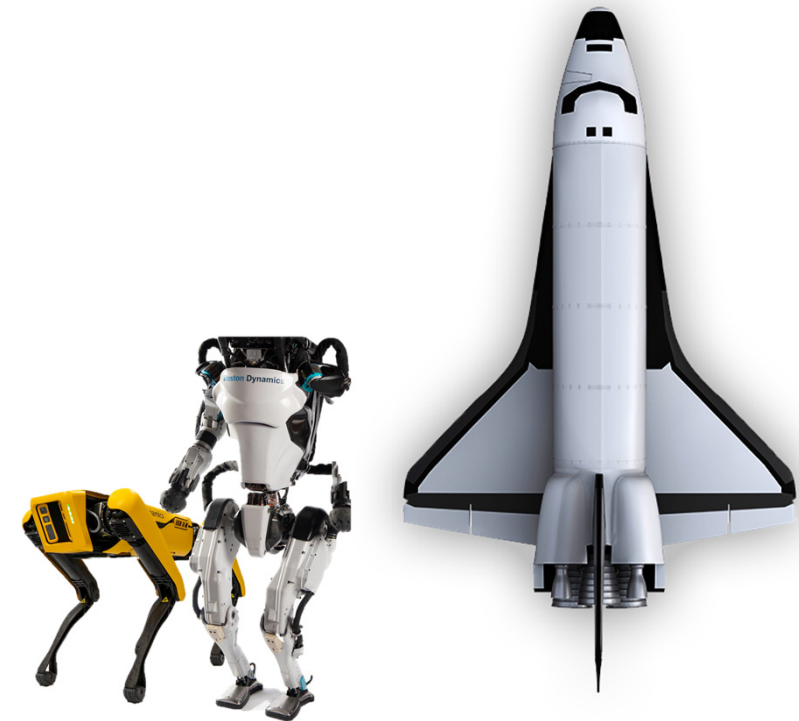
ADVANCED CONTROL SYSTEM DESIGN

Lecture 3

Transformation of State-Space



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Lecture Outline

01

State-Space Representations in Canonical Forms

02

Controllable Canonical Form

03

Observable Canonical Form

04

Diagonal Canonical Form

05

Jordan Canonical Form

06

State-Space Representation to Transfer Function

State-Space Representations in Canonical Forms

Differential Equations ↔ Transfer Functions ↔ State-Space

- There are many techniques for obtaining state-space representations of differential equations and transfer-function systems.
- The system defined by Equation (1) can also be represented using Equation (2):

$$y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} \dot{y} + a_n y = b_0 u^{(n)} + b_1 u^{(n-1)} + \dots + b_{n-1} \dot{u} + b_n u \quad (1)$$

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} \quad (2)$$

- The state-space representation of the system defined in Equations (1) and (2) can be given using the canonical forms.

CCF

Controllable Canonical Form



OCF

Observable Canonical Form



DCF

Diagonal Canonical Form



JCF

Jordan Canonical Form



Controllable Canonical Form

$$y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} \dot{y} + a_n y = b_0 u^{(n)} + b_1 u^{(n-1)} + \dots + b_{n-1} \dot{u} + b_n u \quad (1)$$

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} \quad (2)$$

The controllable canonical form of the system defined in Equations (1) and (2) can be represented using the canonical form below:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \dots & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u$$

$$y = [b_n - a_n b_0 \quad b_{n-1} - a_{n-1} b_0 \quad \dots \quad b_1 - a_1 b_0] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_0 u$$

Controllable Canonical Form

01

The controllable canonical form plays an important role in control theory since it represents the so-called **controllable system**.

02

The controllable canonical form is important in discussing the **pole-placement** approach to control systems design.

03

System **controllability** is one of the main concepts of modern control theory.

Observable Canonical Form

$$y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} \dot{y} + a_n y = b_0 u^{(n)} + b_1 u^{(n-1)} + \dots + b_{n-1} \dot{u} + b_n u \quad (1)$$

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} \quad (2)$$

The observable canonical form of the system defined in Equations (1) and (2) can be represented using the canonical form below:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 0 & \dots & 0 & -a_n \\ 1 & 0 & \dots & 0 & -a_{n-1} \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} b_n - a_n b_0 \\ b_{n-1} - a_{n-1} b_0 \\ \vdots \\ b_1 - a_1 b_0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 0 & \dots & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + b_0 u$$

Observable Canonical Form

01

The observer canonical form is related to another important concept of modern control theory: **system observability**.

02

Observer canonical form has a very simple structure and represents an **observable system**.

03

The observer canonical form is very useful for computer simulation of linear dynamical systems since it allows the effect of the system **initial conditions** to be taken into account.

Diagonal Canonical Form

$$y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1\dot{y} + a_0y = b_0u^{(n)} + b_1u^{(n-1)} + \dots + b_{n-1}\dot{u} + b_nu \quad (1)$$

$$\frac{Y(s)}{U(s)} = \frac{b_0s^n + b_1s^{n-1} + \dots + b_{n-1}s + b_n}{(s + p_1)(s + p_2)\dots(s + p_n)} = b_0 + \frac{c_1}{s + p_1} + \frac{c_2}{s + p_2} + \dots + \frac{c_n}{s + p_n} \quad (3)$$

Consider the transfer-function system defined by Equation (3). Here we consider the case when the denominator's polynomial involves only **distinct roots**. We can represent the system using the canonical form below:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -p_1 & 0 & \dots & 0 \\ 0 & -p_2 & \dots & 0 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & -p_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} c_1 & c_2 & \dots & c_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + b_0u$$

Jordan Canonical Form

$$y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1\dot{y} + a_0y = b_0u^{(n)} + b_1u^{(n-1)} + \dots + b_{n-1}\dot{u} + b_nu \quad (1)$$

$$\frac{Y(s)}{U(s)} = \frac{b_0s^n + b_1s^{n-1} + \dots + b_{n-1}s + b_n}{(s+p_1)^3(s+p_4)(s+p_5)\dots(s+p_n)} = b_0 + \frac{c_1}{(s+p_1)^3} + \frac{c_2}{(s+p_1)^2} + \frac{c_3}{(s+p_1)} + \frac{c_4}{s+p_4} + \dots + \frac{c_n}{s+p_n} \quad (4)$$

Consider the case when the denominator's polynomial of Equation (4) involves **multiple roots**. For this case, the diagonal canonical form must be modified into the Jordan canonical form.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -p_1 & 1 & 0 & 0 & \dots & 0 \\ 0 & -p_1 & 1 & \vdots & & \vdots \\ 0 & 0 & -p_1 & 0 & \dots & 0 \\ \hline 0 & \dots & 0 & -p_4 & & 0 \\ \vdots & & \vdots & & \ddots & \\ 0 & \dots & 0 & 0 & & -p_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} u$$

$$y = [c_1 \quad c_2 \quad \dots \quad c_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + b_0u$$

Diagonal and Jordan Canonical Forms

01

The diagonal and Jordan canonical forms are implemented in the **diagonalization** to show the system eigenvalues, system roots, and other applications.

02

The **diagonal** canonical form is used when the denominator's polynomial involves only **distinct roots**.

03

The **Jordan** canonical form (named after Camille Jordan) is used when the denominator's polynomial involves only **multiple roots**.

Canonical Forms

EXAMPLE 1

Find the state-space representations of the following system using multiple canonical forms:

$$\frac{Y(s)}{U(s)} = \frac{s+3}{s^2+3s+2}$$

$$\frac{Y(s)}{U(s)} = \frac{s+3}{s^2+3s+2} = \frac{b_1s+b_2}{s^2+a_1s+a_2} \quad (1)$$

Controllable Canonical Form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -a_2 & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [b_2 - a_2b_0 \quad b_1 - a_1b_0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + b_0u$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [3 \quad 1] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Equation (1) can also be used to find the Observable Canonical Form, as follows:

Canonical Forms

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & -a_2 \\ 1 & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_2 - a_2 b_0 \\ b_1 - a_1 b_0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + b_0 u$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 3 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

The transfer function in Equation (1) can also be represented as follows using partial fraction expansion:

$$\frac{Y(s)}{U(s)} = b_0 + \frac{c_1}{s + p_1} + \frac{c_2}{s + p_2} = 0 + \frac{2}{s + 1} - \frac{1}{s + 2} \quad (2)$$

The Diagonal Canonical Form can be found based on Equation (2), as follows:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -p_1 & 0 \\ 0 & -p_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} c_1 & c_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + b_0 u$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Canonical Forms

EXERCISE 1

Find the state-space representations of the following system using multiple canonical forms:

$$\frac{Y(s)}{U(s)} = \frac{4s + 2}{s^3 + 4s^2 + 5s + 2}$$

State-Space Representation to Transfer Function

The derivation of a transfer function of a single-input, single-output system from the state-space equations can be carried out using a specific approach.

Consider a system that is represented in state-space as follows:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u \quad (1)$$

$$y = \mathbf{C}\mathbf{x} + Du \quad (2)$$

To derive the transfer function, Laplace transformation should be applied on Equations (1) and (2):

$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}U(s) \quad (3)$$

$$Y(s) = \mathbf{C}\mathbf{X}(s) + DU(s) \quad (4)$$

State-Space Representation to Transfer Function

Since the transfer function was previously defined as the ratio of the Laplace transform of the output to the Laplace transform of the input when the **initial conditions were zero**, then $x(0)$ is set to be zero.

$$s\mathbf{X}(s) - \mathbf{A}\mathbf{X}(s) = \mathbf{B}U(s) \quad (5)$$

$$(s\mathbf{I} - \mathbf{A})\mathbf{X}(s) = \mathbf{B}U(s) \quad (6)$$

$\mathbf{I}$ is the identity matrix or unit matrix, which can be represented as follows:

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \Rightarrow s\mathbf{I} = \begin{bmatrix} s & 0 & 0 & \cdots & 0 \\ 0 & s & 0 & \cdots & 0 \\ 0 & 0 & s & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & s \end{bmatrix} \quad (7)$$

State-Space Representation to Transfer Function

By multiplying both sides of Equation (6) by $(s\mathbf{I} - \mathbf{A})^{-1}$:

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B}U(s) \quad (8)$$

By substituting (8) into (4):

$$Y(s) = [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} + D]U(s) \quad (9)$$

$$G(s) = \frac{Y(s)}{U(s)} = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} + D \quad (10)$$

EXAMPLE 2

Find the transfer function of the following system that is represented by the following state-space equations:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

State-Space Representation to Transfer Function

$$\mathbf{A} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \mathbf{C} = [1 \quad 0 \quad 0] \quad D = 0$$

$$G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + D$$

$$G(s) = [1 \quad 0 \quad 0] \begin{bmatrix} s+1 & -1 & 0 \\ 0 & s+1 & -1 \\ 0 & 0 & s+2 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$G(s) = [1 \quad 0 \quad 0] \begin{bmatrix} \frac{1}{s+1} & \frac{1}{(s+1)^2} & \frac{1}{(s+1)^2(s+2)} \\ 0 & \frac{1}{s+1} & \frac{1}{(s+1)(s+2)} \\ 0 & 0 & \frac{1}{s+2} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$G(s) = \frac{1}{(s+1)^2(s+2)} = \frac{1}{s^3 + 4s^2 + 5s + 2}$$

State-Space Analysis

Next Week ...

**DO YOU
HAVE ANY
QUESTIONS**